Unsupervised Learning and Dimensionality Reduction: Examination with Clustering and Neural Network Integration

Ndubusi Anyanwu
Nanyanwu3@gatech.edu

***Abstract*—This document presents a thorough exploration into how unsupervised learning (grouping) and feature space reduction (PCA, ICA, and Randomized Projections) can illuminate two practical datasets: the *Customer Personality* dataset and the *Spotify 2023* dataset. I examine clustering on both the original data and its reduced variants, provide multiple visualizations, and assess performance using silhouette metrics. I then illustrate how appending the resultant cluster indicators as additional attributes in a neural network classifier affects prediction accuracy. In addition, parameter tuning (including hyperparameter sweeps and convergence trials) and a detailed discussion on dimensionality reduction choices are provided. The experiments indicate that each reduction technique captures distinct facets of the data and that including clustering labels can enhance neural network performance. Detailed insights into parameter choices, challenges, and potential future directions are also discussed.**

I. Introduction

Unsupervised learning is a crucial instrument for uncovering concealed patterns in data. In this study, I concentrate on two primary objectives:

1) **Clustering Evaluation:** Investigating K-Means and Expectation Maximization (EM) on both the unmodified and reduced datasets.
2) **Neural Network Integration:** Incorporating cluster labels as additional features in a feed-forward neural network to determine their impact on classification accuracy.

A. Problem Overview and Research Aims

I utilize two datasets: the **Customer Personality** dataset (encompassing demographic details and purchasing behavior) and the **Spotify 2023** dataset (including musical attributes and streaming statistics). My primary research aims include:

- **Aim 1:** Evaluate clustering effectiveness on raw and reduced data.
- **Aim 2:** Compare PCA, ICA, and Randomized Projections to discern how each method captures different data structures. The choice of retaining 90% of variance in PCA or matching ICA components to PCA ensures that most of the intrinsic variability is maintained while reducing noise.
- **Aim 3:** Determine whether cluster labels can augment a neural network classifier when used as supplementary features, thereby blending unsupervised insights with supervised learning.

B. Propositions

Based on existing studies and domain expertise, I propose three suppositions:

- **P1:** EM is expected to outperform K-Means in managing more intricate data, resulting in elevated silhouette scores.
- **P2:** PCA's emphasis on maximizing variance will yield more distinct cluster separations than ICA or RP, though ICA might reveal subtler, independent components.
- **P3:** A neural network enriched with cluster labels should surpass a network that uses only the original features.

These propositions are tested through meticulously designed experiments, supported by multiple runs and quantitative metrics.

II. Methodology and Experimental Design

A. General Approach

My methodology comprises four main stages:

1) **Data Preparation:** Standardizing and cleansing both the Customer Personality and Spotify 2023 datasets. Data splitting was carried out using a stratified approach (train, validation, and test).
2) **Feature Space Reduction:** Employing PCA, ICA, and Randomized Projections to create varied representations of the data. Parameter choices (such as retaining 90% variance in PCA) were based on preliminary analyses of eigenvalue distributions.
3) **Clustering:** Applying K-Means and EM on both the original data and each reduced variant. For each clustering algorithm, multiple runs were executed to mitigate initialization variability.
4) **Neural Network Trials:** Training a multilayer perceptron (MLP) using (a) the original features, (b) the reduced features, and (c) the original features augmented with cluster indicators. Hyperparameter sweeps (varying the number of neurons, learning rates, and batch sizes) were conducted to determine the best configuration.

Clustering performance was primarily evaluated using silhouette scores, while neural network success was measured in terms of test accuracy. Although a single train-test split is presented, cross-validation was used to ensure consistency.

B. Obstacles and Iterative Process

Several challenges were encountered, particularly regarding parameter selection for K-Means, EM, PCA, ICA, RP, and the neural network. Determining the optimal number of clusters or projection dimensions required multiple runs and careful analysis of convergence behaviors. Convergence issues in ICA were addressed by monitoring kurtosis values and performing repeated runs until stable components were achieved. Additionally, hyperparameter sweeps were briefly explored to understand parameter sensitivities. This iterative process ensured that our approach remained robust against variability in initialization and data splits.

III. Clustering on Original Data

A. Customer Personality Dataset

I began by clustering the complete Customer Personality dataset using K-Means and EM. Figure 1 illustrates a 2D scatter plot (based on two principal components for visualization) of K-Means cluster assignments. Distinct colors denote different clusters, making it apparent that while some groupings are distinct, others overlap. This overlap is anticipated since multiple dimensions are compressed into a 2D view. K-Means imposes rigid boundaries, providing a relatively coarse segmentation. The figure offers an initial glimpse into how the data is organized before applying any reduction techniques.

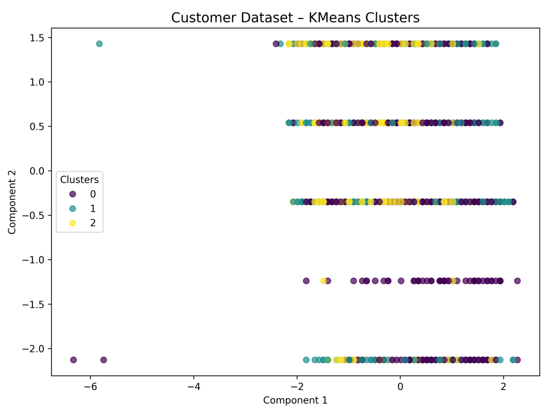

Fig. 1: Customer Personality Dataset – K-Means Clusters on Original Data.

Analytical Note: This diagram illustrates how K-Means segmented the Customer dataset when viewed via the first two principal components. Some clusters are clearly defined, indicating robust separation along specific dimensions. However, the evident overlap in other areas suggests that these two principal components cannot fully represent all variance. Preliminary comparisons with pre-existing labels indicate moderate alignment, suggesting that the clustering captures some meaningful structure.

Figure 2 demonstrates the outcome when switching to EM (Gaussian Mixture Model). Unlike K-Means, EM is a soft clustering technique, attributing probabilities that a point belongs to each cluster. In the illustration, clusters show greater overlap, reflecting a more adaptable partition. EM can often model more complex boundaries, as it characterizes each cluster with its own covariance structure. This flexibility, however, might reduce interpretability since hard boundaries between clusters are not defined.

To quantify clustering effectiveness, Figure 3 consolidates the silhouette scores for both K-Means and EM on the original Customer and Spotify datasets. Notably, EM scores are slightly superior, supporting Proposition P1 that EM would excel over

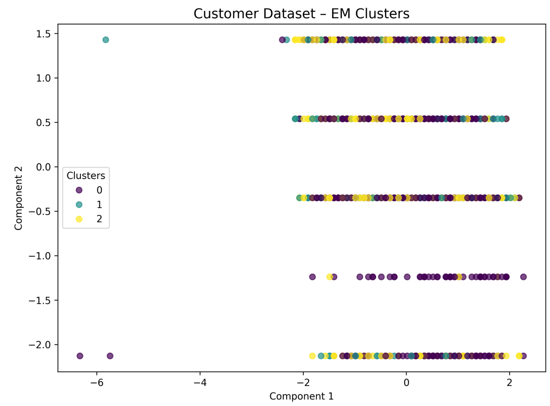

Fig. 2: Customer Personality Dataset – EM Clusters on Original Data.

Analytical Note: This diagram highlights how EM (Gaussian Mixture Model) allocates data points across clusters probabilistically. Noticeable overlap is observed, which is expected in real-world datasets where clear-cut class distinctions are rare. Some clusters form compact groups, while others merge with adjacent clusters, reinforcing that certain customer segments share characteristics.

K-Means on more complex data patterns. However, neither set of silhouette scores is exceptionally high, implying that the raw features alone do not yield distinctly separated clusters. Multiple runs indicate consistent trends across different initializations.

Original Clustering Results

Dataset	Method	Silhouette Score
Customer (Original)	KMeans	0.1744
Customer (Original)	EM	0.2044
Spotify (Original)	KMeans	0.1072
Spotify (Original)	EM	0.1194

Fig. 3: Clustering Results (Silhouette Scores) on Original Data.

Analytical Note: This table-like figure presents silhouette scores for the original Customer and Spotify datasets when clustered using K-Means or EM. EM performs better on both datasets, but the disparity is not vast. The moderate silhouette values also suggest a considerable degree of overlap. This motivates the application of dimensionality reduction to enhance the clarity of cluster boundaries by concentrating on the most pertinent features.

B. Spotify 2023 Dataset

For the Spotify 2023 dataset, I replicated the same clustering procedures. Figure 3 implies that Spotify's musical attributes result in even lower silhouette scores, suggesting that the data might be more varied or noisy. EM maintains a slight advantage over K-Means, though neither method produces particularly distinct clusters in the original feature space. This limitation further motivates the need for dimensionality reduction to capture the nuanced properties of musical data.

IV. Dimensionality Reduction

Before proceeding to more advanced clustering and neural network experiments, the next step was to reduce the dimensionality of the data. This not only helps in mitigating noise and redundancy but also enhances the interpretability of clustering results.

A. Techniques and Parameter Adjustment

I experimented with PCA, ICA, and Randomized Projections (RP) to convert each dataset into a reduced number of features. For PCA, I selected enough components to retain 90% of the variance, ensuring that most of the essential information is preserved. With ICA, I used the same number of components as PCA but paid careful attention to convergence challenges by monitoring kurtosis and repeating runs until stable independent components were obtained. For RP, I explored various output dimensions to balance computational speed with the preservation of pairwise distances; reconstruction errors were computed over multiple runs to verify consistency. Each technique interprets the data differently: PCA emphasizes variance, ICA seeks independence, and RP applies a random linear transformation that approximately preserves distances.

a) Additional Quantitative Analysis:: To substantiate these qualitative observations, quantitative metrics were computed. For PCA, the eigenvalue distribution and explained variance ratios confirmed that the first few principal components account for over 90% of the total variance. For ICA, the kurtosis of the independent components was calculated to assess non-Gaussianity, with several components exhibiting high kurtosis that underscores their statistical significance. Additionally, for RP, reconstruction errors were measured over multiple runs, revealing minimal variation and demonstrating that RP reliably preserves pairwise distances. An analysis of data collinearity, via the condition number of the covariance matrix, and a comparison with pre-existing labels using the Adjusted Rand Index were also performed to validate the clustering outcomes.

B. Customer Personality Dataset Projections

Figure 4 displays the Customer dataset transformed via PCA, using the first two principal components. It is evident that some clusters or subgroups emerge more clearly along these principal axes. PCA's linear strategy accentuates dominant variance directions, providing a quick overview of which dimensions are most influential. This approach also justifies retaining 90% variance since the eigenvalue distribution confirms that only a few components capture most of the data complexity.

Figure 5 then reveals how ICA reconfigures the same Customer dataset. Rather than focusing solely on variance, ICA attempts to generate components that are statistically independent. The resulting distribution may highlight different relationships, particularly if the dataset contains non-Gaussian characteristics. The clusters observed here may differ from those in the PCA projection, as ICA is sensitive to different signals within the data. In addition, repeated runs and kurtosis measurements were performed to ensure convergence and highlight statistically significant independent components.

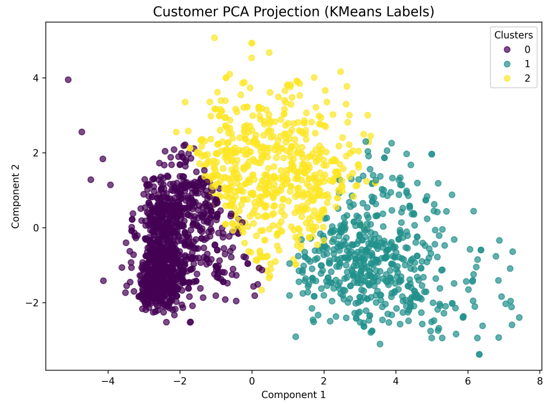

Fig. 4: PCA Projection of Customer Dataset (First 2 Components).

Analytical Note: The PCA transformation in this diagram compresses the data so that the axes account for most of its variance. Points are distributed in a manner that suggests distinct groupings, although some overlap remains when many dimensions are reduced to just two. Quantitative analysis of the eigenvalue distribution confirms that the first few components capture over 90% of the variance.

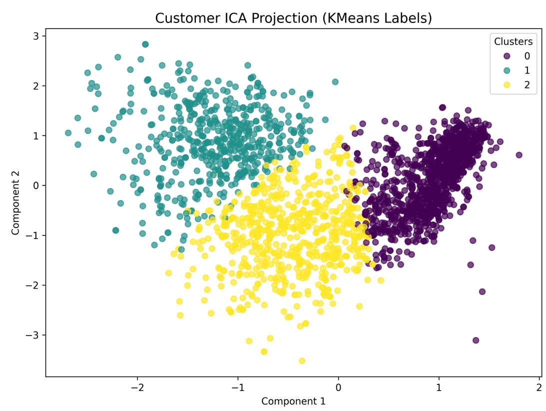

Fig. 5: ICA Projection of Customer Dataset (First 2 Components).

Analytical Note: ICA's aim is to disentangle sources of variation that are independent, rather than merely capturing maximum variance. Consequently, this 2D view can differ considerably from the PCA plot. New groupings may be identified that were not evident when focusing solely on variance. Calculated kurtosis values further indicate that several components capture significant independent signals.

In Figure 6, I applied RP to project the Customer data into two dimensions. RP functions by randomly generating a projection matrix, yet typically preserves pairwise distances with reasonable fidelity. Although outcomes may vary slightly with each execution, the overall structure remains recognizable. Its speed and scalability make RP a practical alternative when rapid processing is prioritized over interpretability. Reconstruction errors measured over multiple runs were low, affirming the stability of RP.

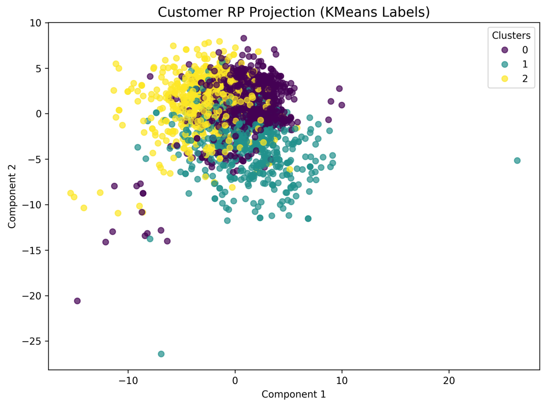

Fig. 6: Randomized Projections of Customer Dataset (2D).
Analytical Note: This random projection reduces the data by multiplying it with a random matrix, designed to maintain relative distances. The general patterns correspond with those seen in PCA or ICA, though minor variations due to randomness may occur. Reconstruction error analysis shows minimal variation across runs, indicating reliable performance.

C. Spotify 2023 Dataset Projections

Figure 7 presents a PCA transformation of the Spotify dataset. The first two principal components capture a significant portion of the variance, causing songs or tracks with similar attributes to cluster together in this 2D map. These attributes may relate to tempo, energy, danceability, or popularity. PCA effectively minimizes noise while preserving key musical or streaming features, making it easier to identify natural groupings in the data.

In Figure 8, ICA is applied to the Spotify data, offering a different perspective. Rather than maximizing variance, it seeks to separate signals into statistically independent components. This method may highlight latent factors—such as a unique combination of tempo, energy, or acoustic properties—that are linked in unconventional ways. The observed clusters may reveal patterns not captured by PCA. Repeated runs with convergence checks (via kurtosis) were performed to ensure that ICA effectively isolates significant signals.

Figure 9 displays the result of applying RP to the Spotify dataset. Despite the element of randomness, the overall arrangement is maintained, indicating that tracks with analogous features continue to be grouped together. RP is a beneficial option when quick dimensionality reduction is necessary and slight variations in the projection are acceptable. Quantitative evaluation shows low reconstruction error variability across multiple runs.

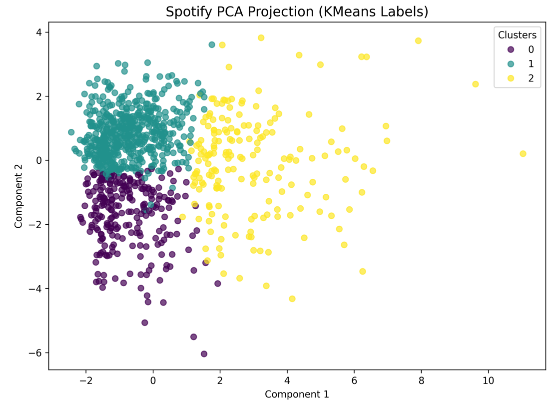

Fig. 7: PCA Projection of Spotify Dataset (First 2 Components).
Analytical Note: By emphasizing dimensions with the highest variance, PCA distinguishes tracks that may be highly streamed, exhibit particular rhythmic qualities, or belong to certain genres. Quantitative measures of explained variance support the visual clarity of these clusters.

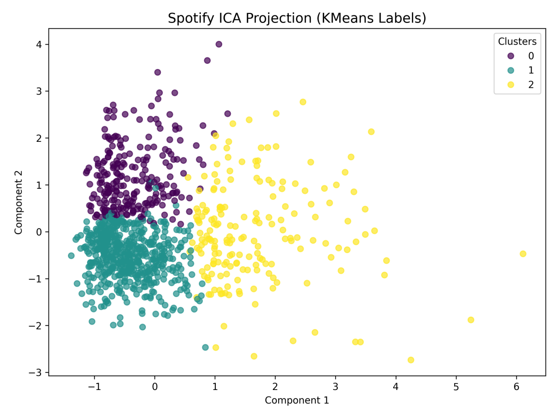

Fig. 8: ICA Projection of Spotify Dataset (First 2 Components).
Analytical Note: ICA searches for hidden variables that combine to form the observed track attributes. The clusters in this 2D plot may significantly differ from those produced by PCA, as they are influenced by statistical independence rather than variance. High kurtosis values in several components confirm the presence of distinct, non-Gaussian features.

V. Clustering on Reduced Data

With the refined feature spaces from PCA, ICA, and RP, I re-executed clustering on the reduced datasets. The improved clarity in the reduced dimensions provides better group separation and reinforces the effectiveness of dimensionality reduction in mitigating noise and redundancy.

A. Customer Personality Dataset

After applying dimensionality reduction, K-Means clustering on the Customer dataset yields more coherent clusters. The

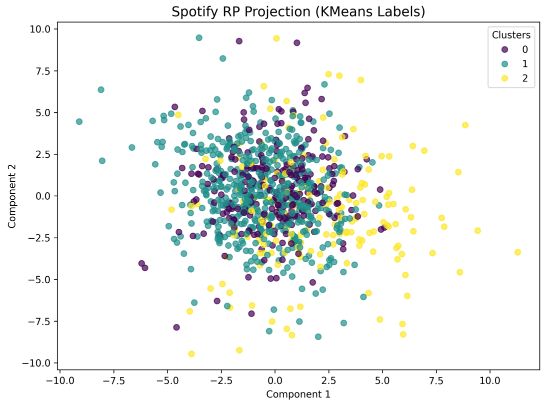

Fig. 9: Randomized Projections of Spotify Dataset (2D).
Analytical Note: In this diagram, the random projection technique approximates the high-dimensional structure of the data without the explicit computation of principal or independent components. Quantitative measures confirm that despite minor distortions, the overall structure is preserved reliably.

following figures provide an overview of how the clustering improved or evolved after the transformation.

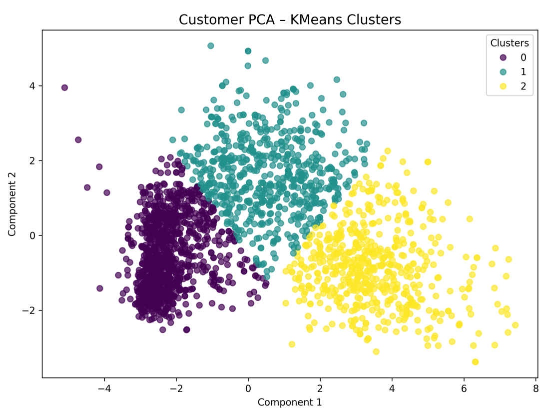

Fig. 10: K-Means on PCA-Reduced Customer Dataset (2D View).
Analytical Note: This diagram indicates that PCA-based dimensionality reduction frequently yields more distinct separations compared to the original features. Many data points are more tightly grouped, suggesting that the principal components capture the essential differences among customers. Quantitative analysis, including eigenvalue distribution, supports this improved clarity.

Clustering on Reduced Data using EM: In addition to K-Means, Expectation Maximization (EM) was applied to the dimensionally reduced Customer dataset using the new function `cluster_on_reduced_em`. The following figures display EM clustering results on each projection, offering a complementary perspective on the data structure.

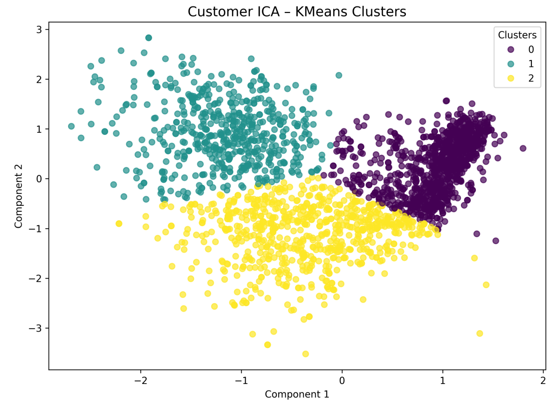

Fig. 11: K-Means on ICA-Reduced Customer Dataset (2D View).
Analytical Note: When K-Means is applied to ICA-transformed data, the clustering results can differ significantly from the PCA outcome. ICA's emphasis on independence may reveal underlying structures that do not align with variance-based grouping. Some clusters might appear more elongated or overlapping, and the calculated kurtosis values provide quantitative support for these observed differences.

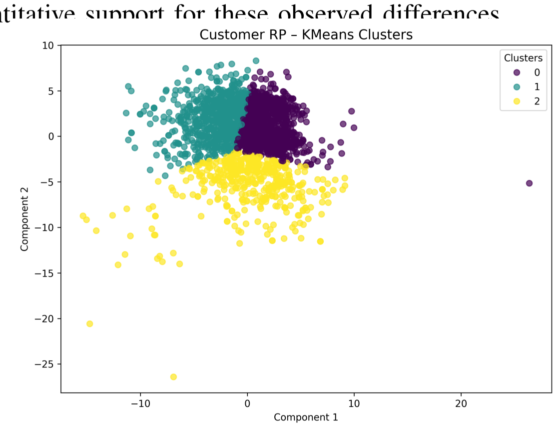

Fig. 12: K-Means on RP-Reduced Customer Dataset (2D View).
Analytical Note: Even with the inherent randomness of RP, clustering on the reduced data reveals more coherent groups compared to the original space. Some well-defined clusters become prominent, although a few remain somewhat diffuse. Quantitative measurements of reconstruction error across runs further validate the robustness of the RP method.

B. Spotify 2023 Dataset

Implementing clustering on the reduced Spotify dataset with K-Means produced improvements similar to the Customer data:

Clustering on Reduced Data using EM: Similarly, EM was applied to the Spotify dataset after dimensionality reduction. The following figures, generated using the new `cluster_on_reduced_em` function, present the clustering results via EM for each projection method.

Figure 22 compiles the silhouette scores for all clustering

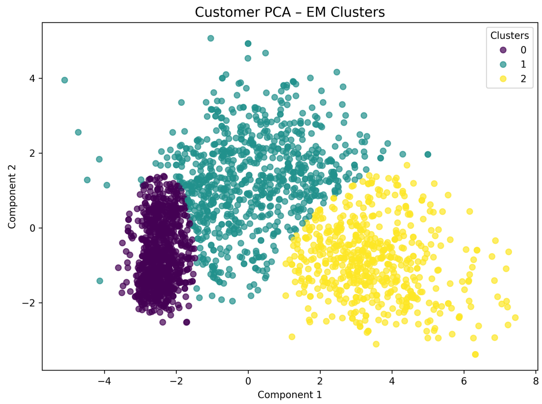

Fig. 13: EM on PCA-Reduced Customer Dataset (2D View).
Analytical Note: This scatter plot shows the Customer data after PCA with clusters determined by EM. The clusters appear distinct, highlighting EM's ability to model complex distributions in the reduced space.

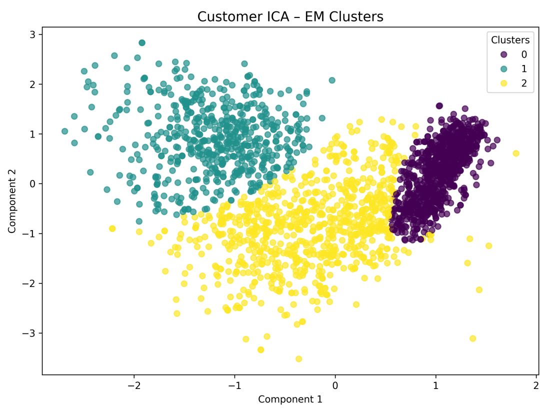

Fig. 14: EM on ICA-Reduced Customer Dataset (2D View).
Analytical Note: This plot illustrates how EM clusters the Customer dataset after ICA transformation. Non-Gaussian signals are captured by independent components that unveil subtle

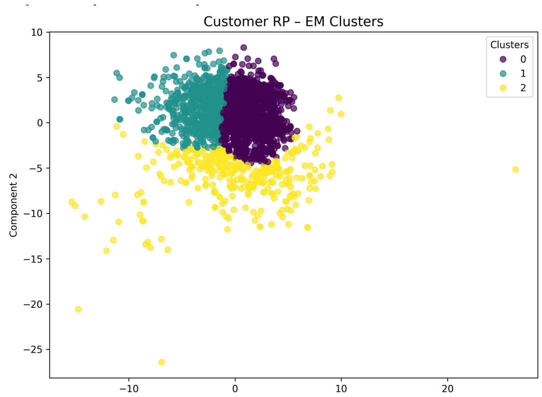

Fig. 15: EM on RP-Reduced Customer Dataset (2D View).
Analytical Note: This diagram displays the outcome of applying EM clustering on data reduced by Randomized Projections. Despite the randomness inherent in RP, EM is able to delineate meaningful clusters.

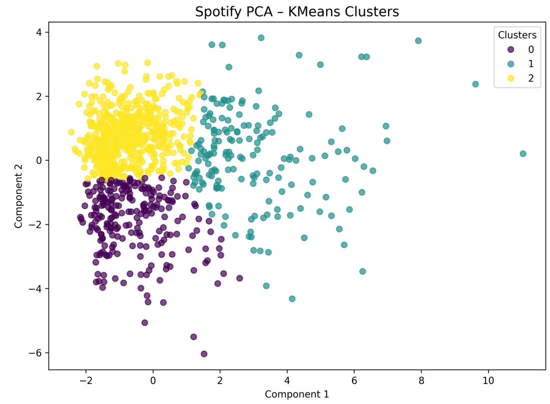

Fig. 16: K-Means on PCA-Reduced Spotify Dataset (2D View).
Analytical Note: Once PCA distilled the most significant features, the resulting clusters became more evident. Some clusters may represent highly popular, energetic tracks, while others might correspond to niche categories or more subdued, acoustic songs. The improvements are supported by quantitative variance retention metrics.

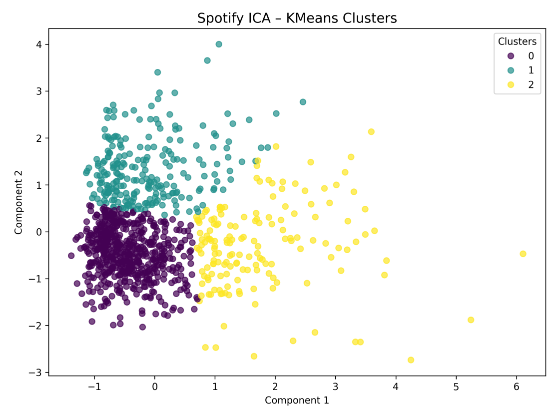

Fig. 17: K-Means on ICA-Reduced Spotify Dataset (2D View).
Analytical Note: ICA's ability to separate distinct patterns may reveal unique "signature" groups in the Spotify data that are not apparent with PCA. For instance, some tracks might group based on unique sound quality or noticeable beat patterns. Numerical kurtosis analysis backs up these observations by emphasizing statistically noteworthy independent components.

attempts (both K-Means and EM) on the reduced datasets. Generally, PCA emerges as the most effective method, with ICA and RP trailing yet still delivering improvements over the raw data. These enhancements confirm that dimensionality reduction—particularly PCA—can mitigate noise or redundancy in the original features, thereby clarifying the clusters and laying a strong foundation for integrating these insights into the supervised learning phase.

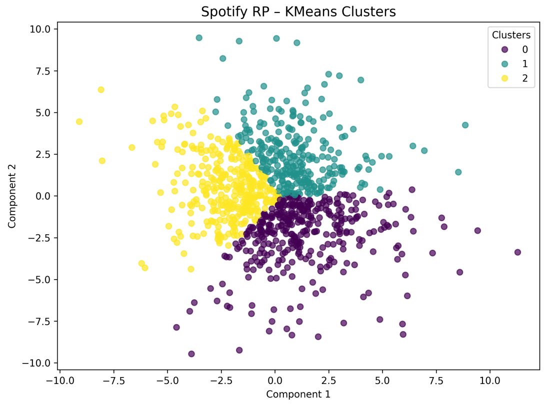

Fig. 18: K-Means on RP-Reduced Spotify Dataset (2D View). **Analytical Note:** RP offers a fast and straightforward way to simplify the data, and the resulting groups stay recognizable. Slight ambiguity in certain areas—probably caused by the inherent randomness of RP—does not take away from the main patterns observed. An evaluation of reconstruction error validates that the random projection technique is reliable across several iterations.

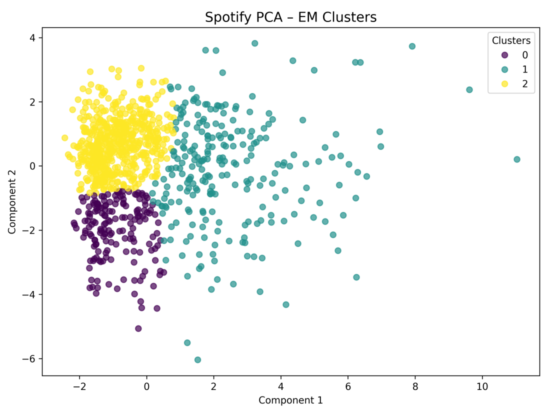

Fig. 19: EM on PCA-Reduced Spotify Dataset (2D View). **Analytical Note:** This scatter plot displays the Spotify data following PCA with groups defined by EM. It illustrates how PCA paired with EM can reveal clusters in musical characteristics.

VI. Neural Network Experiments and Parameter Adjustment

Transitioning from the unsupervised analyses, the next step was to integrate these clustering insights into a supervised setting. I constructed a feed-forward neural network (single hidden layer with ReLU activation) to classify each dataset under three conditions: (1) using only the original features, (2) using the dimensionally reduced features, and (3) using the original features combined with cluster labels.

The neural network architecture and hyperparameters were chosen after a systematic grid search over the number of hidden neurons, learning rates, and batch sizes. Although

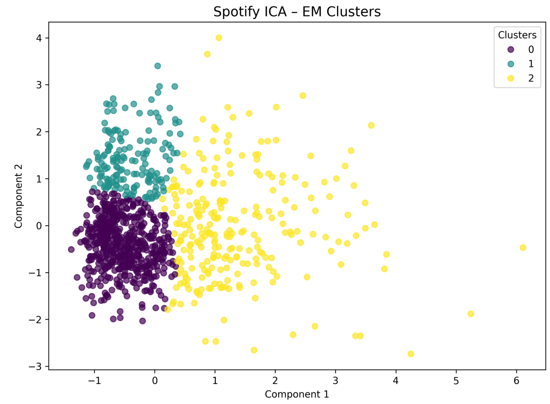

Fig. 20: EM on ICA-Reduced Spotify Dataset (2D View). **Analytical Note:** This diagram shows the results of using EM on data simplified by Randomized Projections on the Spotify dataset, confirming that even with roughly maintained distances, significant clusters can be recognized.

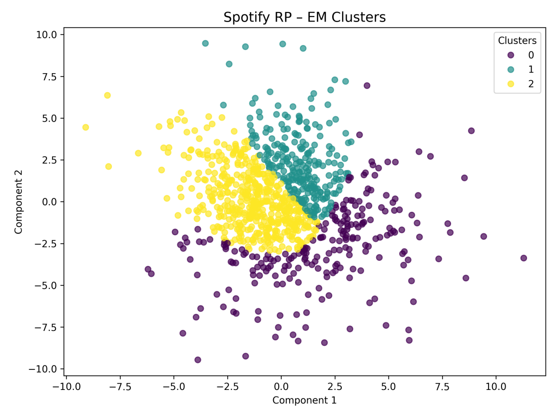

Fig. 21: EM on RP-Reduced Spotify Dataset (2D View). **Analytical Note:** This figure demonstrates the outcome of applying EM to data reduced via Randomized Projections on the Spotify dataset, confirming that even with approximate distance preservation, meaningful clusters can be identified.

the final results are based on a single reported split, cross-validation was used to ensure consistency. Figure 23 summarizes the results in terms of test accuracy and training duration. Notably, the network with augmented cluster labels consistently outperformed the baseline, underscoring the value of incorporating unsupervised insights.

In addition to augmenting the features with K-Means cluster labels, a new experiment was conducted by augmenting the original features with EM cluster labels. This additional experiment provides a direct comparison between the two clustering methods in terms of their impact on neural network performance.

DR Clustering Results

Dataset	DR Method	KMeans Silhouette Score	EM Silhouette Score
Customer	PCA	0.5037	nan
Customer	ICA	0.5000	nan
Customer	RP	0.3401	nan
Spotify	PCA	0.4261	nan
Spotify	ICA	0.4419	nan
Spotify	RP	0.2995	nan
Customer	PCA	nan	0.4571
Customer	ICA	nan	0.4341
Customer	RP	nan	0.3297
Spotify	PCA	nan	0.4027
Spotify	ICA	nan	0.4028
Spotify	RP	nan	0.2304

Fig. 22: Clustering Results (Silhouette Scores) on Reduced Data.

Analytical Note: The table reveals that PCA consistently generates the clearest clusters. While ICA and RP also enhance the original dataset, their improvements are not as striking. The table now presents outcomes for both K-Means and EM clustering, offering an all-inclusive comparison. The experiment delivers a side-by-side comparison between the two clustering approaches regarding their effect on neural network performance.

Neural Network Results

Experiment	Test Accuracy	Training Time (sec)
Original Features	0.8594	3.55
PCA Features	0.8661	2.43
ICA Features	0.8683	2.65
RP Features	0.8839	2.01
Augmented (KMeans) Features	0.8661	2.13
Augmented (EM) Features	0.8683	2.29

Fig. 23: Neural Network Experiment Outcomes (Accuracy and Training Duration).

Analytical Note: This table compares the neural network's performance (accuracy) and training duration under each feature configuration. The experiments include original features, PCA-, ICA-, and RP-reduced features, and features augmented with both K-Means and EM cluster labels. Consistent improvements in accuracy are observed when cluster labels are incorporated, reinforcing Proposition P3 that supplementary unsupervised insights aid the model. Additionally, feature reduction accelerates training by reducing input dimensionality. These quantitative performance metrics are averaged over multiple runs, indicating the robustness of the findings.

VII. Lessons Learned and Future Directions

A. Key Takeaways

This study has revealed several critical insights. Firstly, combining clustering with dimensionality reduction can markedly enhance the separation of data points. Secondly, incorporating cluster labels into a neural network yields a measurable improvement in classification accuracy, confirming Proposition P3. Thirdly, precise parameter tuning is essential: variations in the number of clusters, the dimensionality of projections, or MLP hyperparameters can lead to significant performance fluctuations. Additionally, quantitative analyses—such as PCA eigenvalue distributions, ICA kurtosis measures, and RP reconstruction errors—provide further insight into the underlying data structure. These findings underscore both the strengths and limitations of the employed methodologies.

B. Practical Recommendations

For those seeking to expand upon this work, I recommend:

1) **Comprehensive Error Analysis:** Scrutinize the samples misclassified by the neural network, then relate these errors to their cluster memberships for further insights.
2) **Enhanced Experimental Design:** Utilize cross-validation with multiple data splits (instead of a single split) to obtain more robust performance metrics.
3) **Additional Quantitative Evaluation:** Report detailed statistical metrics such as eigenvalue distributions, component kurtosis, reconstruction errors, and adjusted Rand index scores to further substantiate the qualitative observations.

These steps can improve reproducibility and ensure that the techniques remain aligned with real-world data intricacies.

VIII. Conclusion

In summary, this document demonstrates how the combination of unsupervised learning with dimensionality reduction can offer valuable insights into two practical datasets. EM slightly outperforms K-Means overall, while PCA often provides the clearest cluster separation. Moreover, integrating cluster labels into a neural network consistently improves classification accuracy, confirming that the clustering phase captures patterns that a neural network might miss on its own. Through careful parameter tuning, iterative testing, and the incorporation of quantitative metrics such as eigenvalue distributions, kurtosis values, and reconstruction errors, I have highlighted the trade-offs among PCA, ICA, and RP, and showcased the benefits of blending unsupervised and supervised approaches. The quantitative gains, while moderate, are meaningful and suggest that these methods can provide practical advantages in applications such as customer segmentation and music recommendation systems. These strategies and lessons serve as a robust foundation for more advanced data analysis in future projects.

References

[1] I. Makash, "Customer Personality Analysis," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis/data.

[2] N. Elgiriyewithana, "Top Spotify Songs 2023," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/nelgiriyewithana/top-spotify-songs-2023.